

Impact of Iron Availability on the Motility of Various *Escherichia coli* CFT073 Mutants

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Introduction

Urinary tract infections (UTIs) are the second most common type of bacterial infection, affecting primarily women. Approximately 50% of women will contract a urinary tract infection (UTI) at some point in their lives. UTIs are commonly caused by uropathogenic *Escherichia coli* (UPEC) and involve bacterial motility as a virulence factor. Swimming and swarming are flagella-driven types of motility, with swarming being collective movement on solid surfaces. A $\Delta tolC$ mutant of the model uropathogenic strain CFT073 lacking the main outer efflux pump loses the ability to swarm but retains the ability to swim. TonB, part of the TonB-ExbBD complex, energizes iron uptake via outer membrane receptors. Interestingly, the double mutant CFT073 $\Delta tolC/\Delta tonB$ can swim and swarm, like the WT strain. Given that iron deprivation increases swimming motility and lacks the ability to import iron, we hypothesized that iron limitation influences swarming motility in UPEC. Understanding the relationship between iron limitation and the degree of swarming motility may allow us to invent novel treatment methods for UTIs.

Methods

- E. coli* CFT073 WT, $\Delta tolC$, and $\Delta tolC/\Delta tonB$ were incubated overnight in TSB at 37°C.
- Swarming agar plates were poured and left covered at room temperature overnight.
 - Appropriate concentrations of Fe and DIP were added before pouring
- OD was standardized (OD600=1), and 5µL of cells were inoculated on Eiken agar plates.
- Swarming plates were incubated at 37 degrees for 24 hours.
- Images of plates were taken, and swarming areas were measured using ImageJ software.

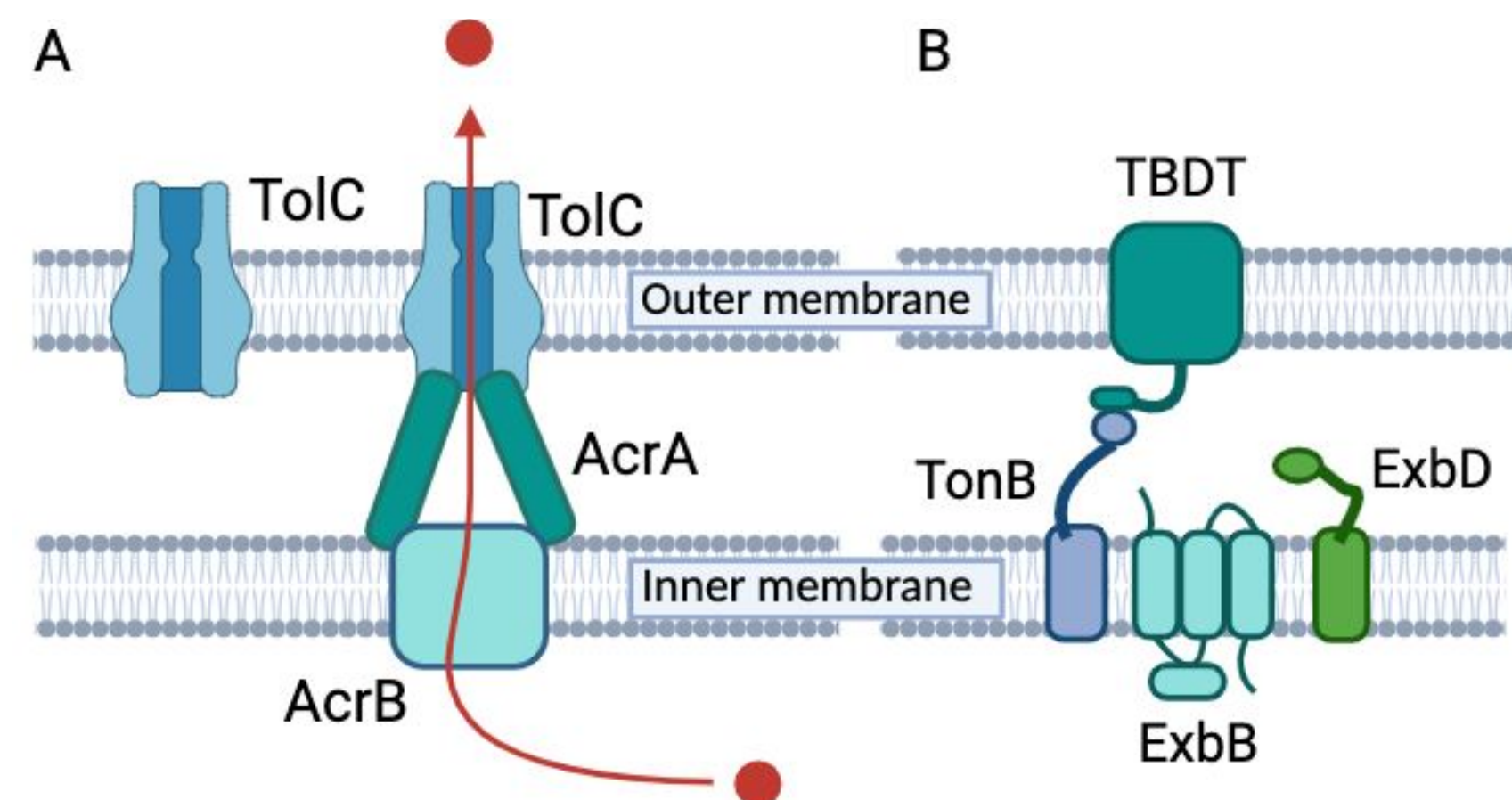


Figure 1. Relevant membrane protein complexes are depicted A) the TolC efflux pump with associated proteins AcrA and AcrB and B) the TonB-ExbBD complex. Created with BioRender.com

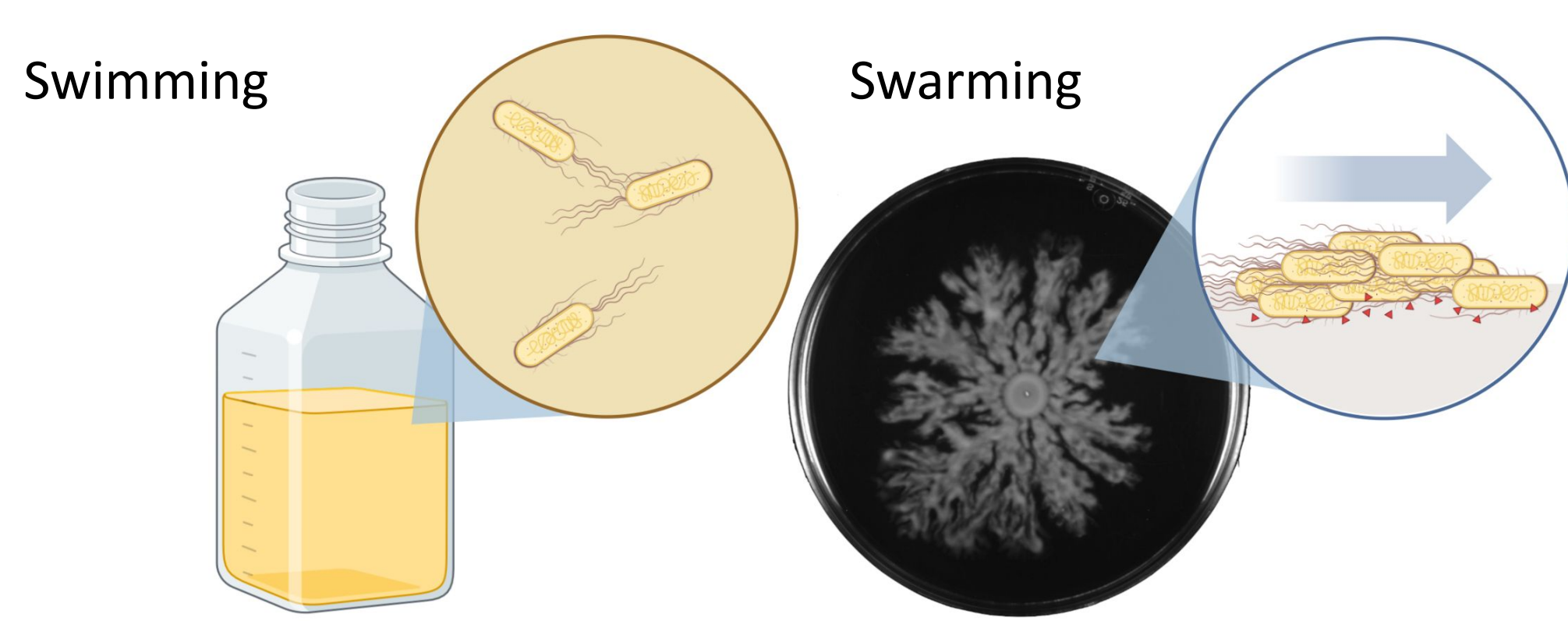


Figure 2. While individual WT *E. coli* expresses swimming motility in liquid media (shown on the left), multicellular swarming motility is expressed in solid media (shown on the right). Swarming often occurs in coordinated cell groups known as rafts, with many bacterial species secreting surfactants (red triangles) that lower the surface tension between the substrate and the cell allowing for successful spreading. This image was created with Biorender.com.

Results

To uncover the role of iron availability in motility, we tested different strains of uropathogenic *E. coli* with differing concentrations of added iron and dipyrldyl (an iron chelator) and measured the subsequent swarming areas.

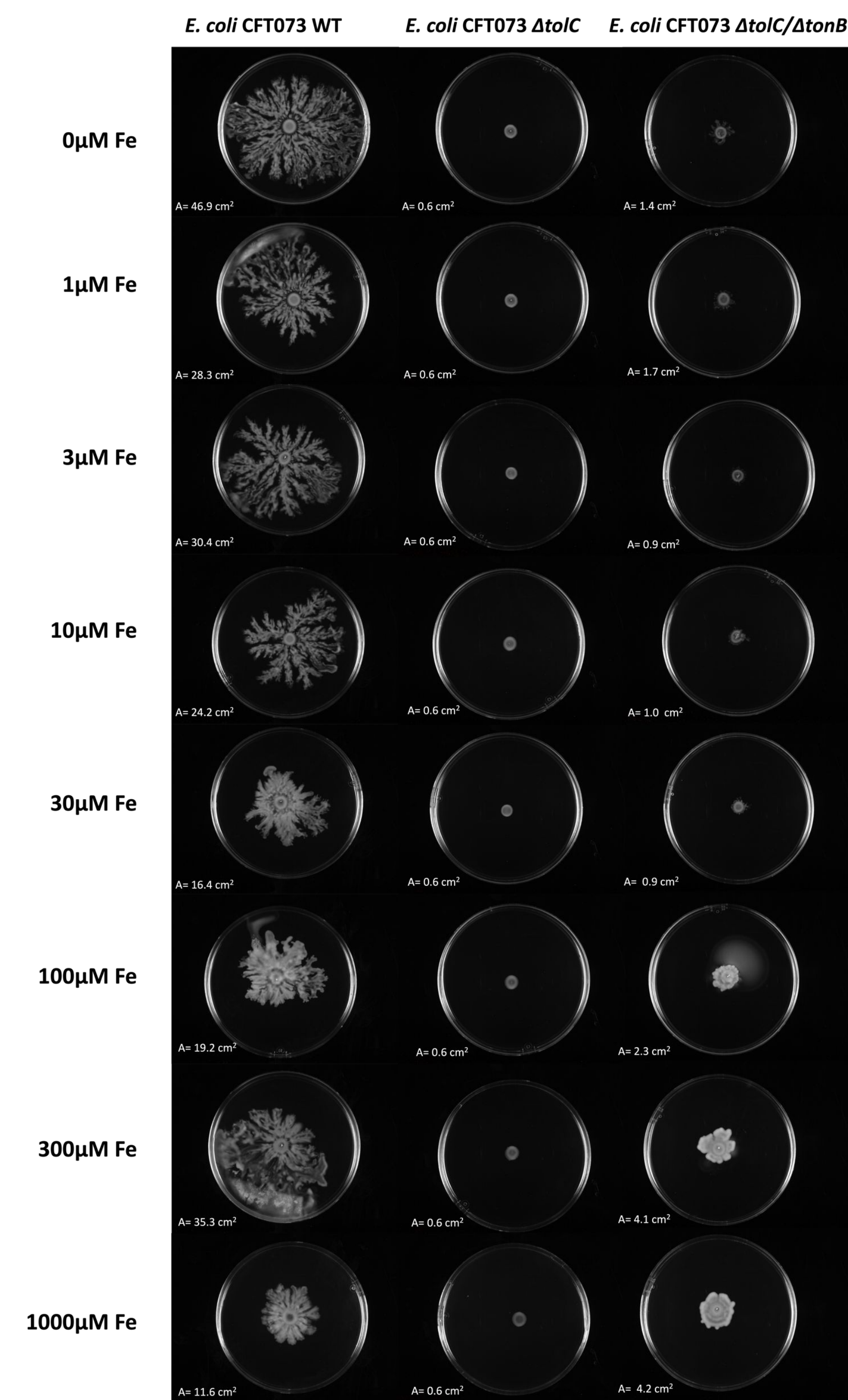


Figure 3. *E. coli* CFT073 WT, $\Delta tolC$, and $\Delta tolC/\Delta tonB$ after 24 hours of incubation at 37 degrees Celsius on swarming agar with various concentrations of added iron.

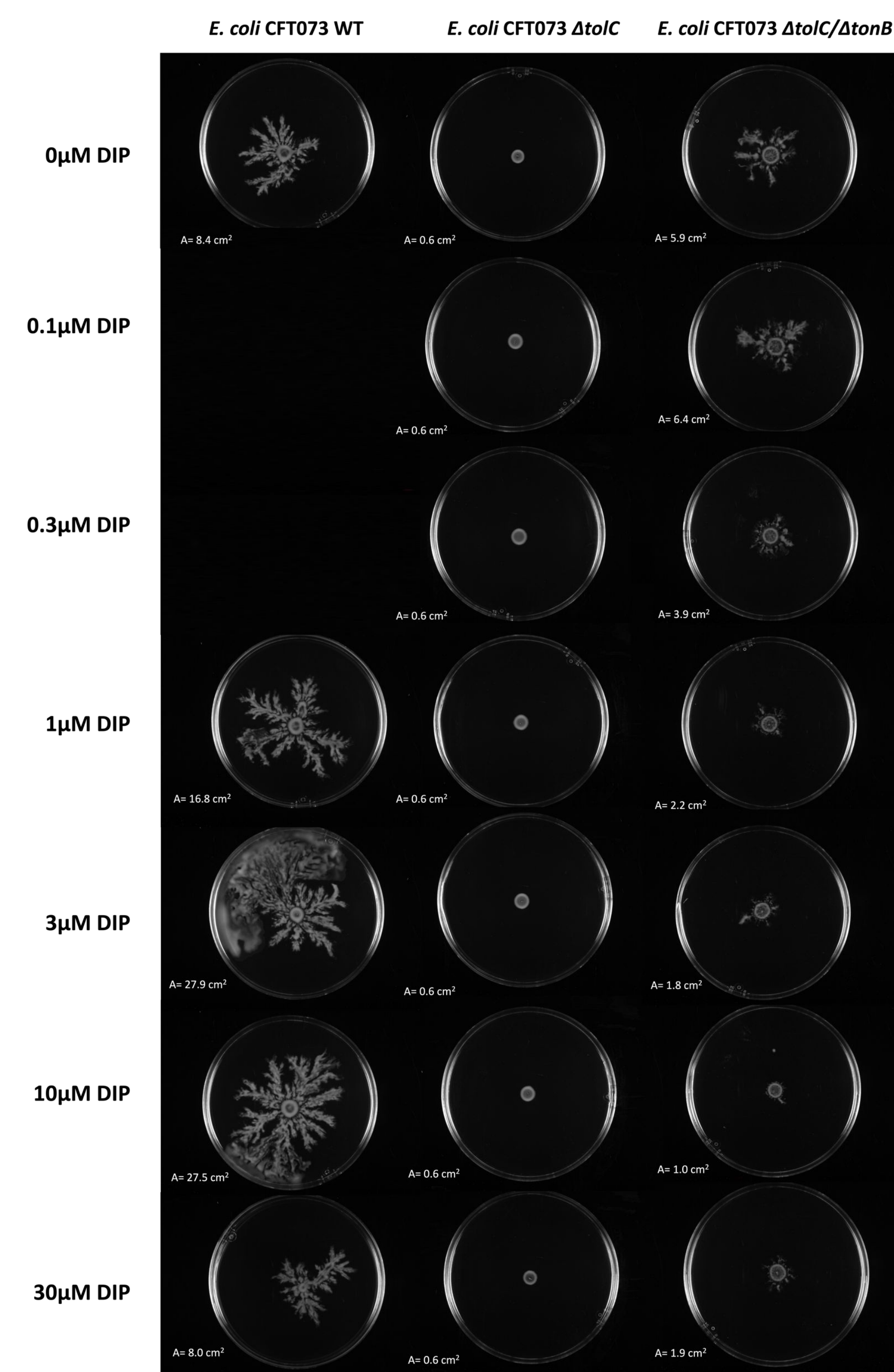


Figure 4. *E. coli* CFT073 WT, $\Delta tolC$, and $\Delta tolC/\Delta tonB$ after 24 hours of incubation at 37 degrees Celsius on swarming agar with various concentrations of dipyrldyl, an iron chelator.

Conclusion

We found that added concentrations of iron tend to decrease the swarming area of wild type *E. coli* CFT073, and that increased concentrations of an iron chelator, dipyrldyl, tends to increase swarming area. However, DIP arrests all growth at concentrations greater than 100 µM. The $\Delta tolC$ mutant showed no change in phenotype across all tested conditions. This led us to the conclusion that iron availability is most likely not the cause of the mutant's inability to swarm. The double mutant showed little variation across conditions, however, at high concentrations of iron, the phenotype visually changed to a more opaque, flowering swarming patter. We observed a similar phenotype in WT at high concentrations of iron. We are still testing other mutants under these conditions and are moving forward by exploring other potential reasons for the $\Delta tolC$ phenotype.

Acknowledgements

Thank you to the William and Linda Frost Fund for its continued support of BCSM undergraduate research and to the Center for Applications in Biotechnology for providing us the facilities to carry out this research.

References

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